RESEARCH COLLABORATION & EXPERT RECOMMENDATION SYSTEM

1. Abstract

In today's academic environment, research scholars face significant challenges in identifying relevant research papers and finding suitable experts for collaboration. Existing platforms provide access to research papers but lack intelligent mechanisms to recommend domain-specific experts for collaboration.

The Research Collaboration & Expert Recommendation System aims to bridge this gap by providing a unified platform that helps users search research papers and recommends experts based on research interests and expertise. The system analyzes keywords and domains to suggest suitable collaborators, thereby saving time and improving research productivity. The proposed system is designed using Object Oriented Analysis and Design principles and implemented using Java.

2. Problem Statement

Research scholars often spend a considerable amount of time searching for relevant research papers and identifying experts working in similar domains. Even after finding useful papers, discovering the right person to collaborate with remains a difficult and time-consuming task. Existing systems focus mainly on paper retrieval and do not support expert recommendation and collaboration effectively. Hence, there is a need for a system that not only provides research paper information but also recommends suitable experts to facilitate collaboration.

3. Objectives

The main objectives of the proposed system are:

- To provide an efficient platform for searching research papers
- To recommend domain-specific experts based on research interests
- To support collaboration between research scholars and experts
- To reduce time and effort involved in finding suitable collaborators
- To demonstrate Object Oriented Analysis and Design concepts

4. Scope of the Project

The scope of this project is limited to building a prototype system that supports basic research paper management, expert profiling, and collaboration recommendations. The system focuses on keyword-based expert recommendation and does not involve real-time data extraction from external research platforms. The project is developed as a mini-project to demonstrate OOAD principles and object-oriented programming concepts.

5. Proposed System

The proposed system allows users to register and log in as research scholars or experts. Research scholars can search for research papers using keywords and view recommended experts based on domain similarity. Experts can maintain their profiles with specialization details. The system matches research keywords with expert expertise to generate recommendations. Users can send collaboration requests, and experts can accept or reject them. The system follows an object-oriented approach with clearly defined classes and relationships.

6. Functional Requirements

- User registration and authentication
- Research paper search by keywords
- Expert profile management
- Expert recommendation based on domain matching
- Sending and managing collaboration requests

7. Non-Functional Requirements

- User-friendly interface
- Secure access to user data
- Scalability for adding more users and papers
- Maintainability using object-oriented design

8. Technologies Used

- Programming Language: Java
- Design Methodology: Object Oriented Analysis and Design (OOAD)
- Database: MySQL / SQLite
- UI: Java Swing / Console-based Interface

9. Conclusion

The Research Collaboration & Expert Recommendation System provides an effective solution for research scholars to find relevant papers and suitable collaborators. By applying object-oriented principles and OOAD methodology, the system ensures modularity, scalability, and ease of maintenance. The project demonstrates how object-oriented design can be used to solve real-world academic problems efficiently.